



FDK™ Toolkit, Hiring Domain Fairness Audit

External Validation Report: HR Analytics: Employee Promotion Data (WNS Analytics Wizard)

Fifth Real-World Benchmark Validation, Following Justice/COMPAS, Education/Bar Passage, Finance/German Credit & HMDA (Washington 2025), Health/UCI Diabetes

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Reference implementation validation for the FDK Hiring domain, extending the no-code UX pattern established during Finance and Health domain validation.

Abstract

This report documents the ground-up development and real-dataset validation of the FDK™ (Fairness Diagnostic Kit) Hiring domain's no-code user experience, following the methodology established during Finance and Health domain validation. As with Health, Hiring's no-code dropdown UX did not exist before this validation cycle; it was built during this work, alongside a fallback-detection fix and a new pipeline-level binary-validation safeguard. Hiring also introduces a feature not present in either prior domain: comparative-study mode, which enforces target-column consistency across paired pre-/post-implementation audits. Validation was conducted against the WNS Analytics Wizard employee promotion dataset, a genuinely real (not synthetic) HR dataset drawn from an actual company's promotion decisions. Two independent group cuts were audited: gender and age, on the same underlying employee population and toy prediction model. Results replicate the pattern found during both HMDA and Diabetes validation: one protected attribute (gender) showed LOW_BIAS while another (age) showed HIGH_BIAS on the identical dataset and model, now confirmed on a third independent dataset in a third domain.

Background

Following Health's establishment as a second working reference implementation alongside Finance, Hiring was audited next, one file at a time rather than all five at once, to reduce the risk of the reviewer working from an inconsistent or hallucinated picture of the codebase. The audit found Hiring in a similar starting position to Health: no dropdown-based override UX existed on either the upload page or the confirmation page, and the same class of fallback-detection bug was present (though in a more permissive form, Hiring's fallback accepted any two-valued column with no value-set check at all, not even Health's {1,2} allowance). Hiring also has a materially different feature neither Finance nor Health has: comparative-study mode, which registers a pre-implementation test's target column and checks that a corresponding post-implementation test uses the same target, for valid before/after comparison. This required care when building the override UX, specifically, deciding whether a user should be allowed to override the target column during an active comparative study.

Dataset Description

HR Analytics: Employee Promotion Data (WNS Analytics Wizard)

- Source: Kaggle (freely downloadable, no application required), originally released as the dataset for the WNS Analytics Wizard hackathon

- Genuinely real HR data, not synthetic, based on an actual company's real-life promotion-decision process, in contrast to several other publicly available "hiring bias" datasets that are explicitly synthetic or programmatically generated (e.g., IBM HR Analytics, and a Kaggle "Recruitment Bias & Fairness AI" dataset generated via probabilistic simulation)
- 54,808 employee records (train.csv only; the accompanying test.csv has no is_promoted labels and was not used), 13 columns including department, region, education, gender, age, recruitment channel, training history, and performance ratings

Preprocessing: age was bucketed into four groups (Under 30, 30-39, 40-49, 50+) to provide a second group cut alongside gender, since age is itself an EEOC-protected class. Note this dataset does not include race/ethnicity as a column; only gender and age are available as genuine protected attributes; department and region exist as columns but are not legally protected classes and were not used as primary group cuts. Minor nulls in education (2,409 rows) and previous_year_rating (4,124 rows) were imputed (mode and median, respectively).

Derived prediction column: as with every dataset used in this project so far, this dataset contains no native model-decision field. A toy LogisticRegression classifier was trained (features = all other columns excluding the identifier, one-hot encoded, including protected attributes as inputs), with class_weight='balanced' applied from the start this time, applying the lesson learned from the Diabetes dataset's initial degenerate fit directly, rather than needing a second corrective pass. Result: 76.3% test accuracy with 67.0% recall on the positive class (is_promoted is imbalanced at ~8.5% positive), non-degenerate predicted-positive counts across every subgroup in both cuts.

Findings: No-Code UX Construction & Bug Remediation

UX Construction (new, not a port)

As with Health, no dropdown-based override UX existed for Hiring before this work. The same three functions (_describe_column, _candidate_columns, build_column_options) were built following the established pattern, with one deliberate domain-specific difference: Hiring's _candidate_columns enforces strict {0,1}-only values for the target/prediction roles, matching Hiring's own primary detection convention, rather than Health's more permissive {1,2}-inclusive clinical-coding convention. auto_confirm_hiring.html was rebuilt from a static read-only display into four real, filtered, pre-selected dropdowns, with the same disabled-run-button safeguard used in Health when no valid target or prediction column exists.

Finding 1, Fallback binary-detection gap (more permissive than Health's equivalent bug)

Root cause: the y_true/y_pred fallback detection paths accepted any two-valued numeric column with no check on the values themselves, not even Health's {1,2} allowance, just a bare `nunique() == 2`. Hiring's primary (keyword-matched) detection already correctly required strict {0,1} values.

Remediation: both fallback paths now require the same strict {0,1} check already used in the primary detection path, deliberately not copying Health's {1,2} leniency, since Hiring's own convention never included it.

Verification: confirmed with a synthetic {1,2}-coded column, which the unpatched fallback would have accepted but the patched version correctly rejects, matching Hiring's own stricter convention.

Finding 2, Layer-3 pipeline validation gap

Root cause: `fdk_hiring_pipeline.py`'s `calculate_all_metrics` function checked required columns, group count, and the 20-sample statistical-power floor, but never independently verified that y_true/y_pred were genuinely binary, the same gap found and fixed in Health.

Remediation: added an independent binary-validation check using the same ValueError-raising style already used by this file's existing checks, keeping strict {0,1} only, consistent with Finding 1.

Verification: confirmed via direct functional testing that a synthetic non-binary two-valued column is correctly rejected, while genuine {0,1} data continues to pass.

Finding 3, Confirmation-page override inputs were already present but silently dead

Root cause: identical pattern to Health, `auto_confirm_hiring.html` already submitted hidden fields (`group_col`, `y_true_col`, `y_pred_col`, `y_prob_col`), but `run_hiring_audit_with_mapping` never read `request.args`, so any override was silently discarded.

Remediation: the four hidden inputs were converted into real, filtered dropdowns, and the run-audit route now reads and re-validates each override against its role's own candidate list before acceptance.

Design decision, comparative-study interaction: overriding the target column during an active comparative study is deliberately still allowed rather than blocked outright, since the pipeline's existing pre-/post-consistency check already detects and warns when the resulting target diverges from the registered pre-test target. A visible warning was added to the confirmation page's target dropdown specifically when a comparative study is active, so a user overriding it is making an informed choice rather than being surprised by an inconsistency warning after the fact.

Bonus fix: the confirmation page template received `comparative_study_id`, `is_comparative`, and `comparative_status` from the backend but never displayed any of it; comparative-study status was invisible to the user even when active. This is now displayed on the confirmation page.

Finding 4, Upload page had no manual target override

Root cause and remediation: identical to Health's equivalent finding, a hidden input auto-filled from the shared `/api/detect-columns` endpoint, with no visible override. Converted to a visible dropdown populated by parsing the chosen file's header row client-side, independent of the shared endpoint's response shape.

Finding 5, Deployment gap during rollout (caught and corrected same-day)

What happened: after the four Hiring files were updated, the first live confirmation-page test failed with `"column_options' is undefined"`; the template-side change (which expects a `column_options` variable) had been applied, but the corresponding backend change that computes and passes `column_options` into the `render_template` call had not yet been applied to the live file.

Detection and resolution: diagnosed from the error text alone (Jinja2's "is undefined" wording specifically indicates a variable never passed at all, as distinct from a missing dict key), pointing directly at the specific find/replace change responsible. Applying that change resolved the error, confirmed by the first fully successful Hiring audit run immediately afterward.

Fairness Findings

Employee Promotion, gender

Group	n	Selection rate	Base rate (actual promotion rate)
Male	38,496	0.2630	(base rates not itemized by group in this report)
Female	16,312	0.2886	(base rates not itemized by group in this report)

- Statistical parity difference: 0.0256
- Disparate impact: 0.9113, comfortably clear of the 0.80 threshold
- Composite bias score: 0.0189, overall assessment: LOW_BIAS ("Generally fair across applicant groups")

Employee Promotion, age

Group	n	Selection rate
30-39	28,030	0.2970
Under 30	14,453	0.2813
40-49	8,952	0.2195
50+	3,373	0.1405

- Statistical parity difference: 0.1565 (30-39 selection rate 0.2970 vs. 50+ 0.1405)
- Disparate impact: 0.4731, well below the 0.80 threshold
- Composite bias score: 0.1394, overall assessment: HIGH_BIAS, flagged by the pipeline as an "employment law compliance concern"

Notable pattern: the 50+ age group's promotion selection rate (14.1%) is less than half the 30-39 group's (29.7%), age is an EEOC-protected class in the United States, and a disparity of this magnitude is exactly the shape of finding an age-discrimination audit is designed to surface.

Cross-domain observation

The same Employee Promotion population and toy model that showed LOW_BIAS on the gender cut (composite bias score 0.0189, disparate impact 0.911) showed HIGH_BIAS on the age cut (composite bias score 0.1394, disparate impact 0.473). This is the third independent replication of the same pattern first found in HMDA (race vs. sex) and repeated in Diabetes (race vs. gender): auditing only one protected attribute on a given dataset does not license a general conclusion about that dataset or model's fairness. Across all three domains now validated, the specific attribute that showed the larger disparity has differed each time (race, race, age), reinforcing that there is no reliable way to predict which attribute will surface the largest disparity without actually testing each one.

Scope Limitations

- The dataset contains no native model-decision column; a toy LogisticRegression classifier was trained to derive predicted outcomes. These fairness findings describe the toy model's behavior, not any real deployed promotion-recommendation system; they validate that FDK's pipeline correctly detects and quantifies disparities, not that WNS or any specific real-world employer exhibits these disparities today.
- This dataset does not include race or ethnicity as a column; only gender and age were available as genuine protected attributes for this validation. A hiring/EEO-context fairness audit that omits race is incomplete by

definition; this limitation is a property of the dataset, not of FDK's capability, but should not be read as a full EEOC-compliance audit of this employer's practices.

- The Hiring confirmation-page dropdown UX, upload-page manual override, and comparative-study-aware warning did not exist before this validation cycle and are new construction, verified against one real dataset so far, not the breadth of edge cases Finance has now seen.
- A deployment gap (Finding 5) occurred when one of four coordinated file edits was applied without the others; this was caught immediately by the first live test rather than persisting undetected, but underscores that multi-file changes in this project depend on every file actually being updated together.
- The `comparative_target_consistent` metadata field reads false in both reports despite both being first-of-their-kind pre-implementation runs with no actual inconsistency; this reflects a pre-existing quirk in the comparative-study logic's fallback path (see Findings) rather than a real target mismatch, and should not be read as a data quality problem.

Conclusion

The FDK Hiring domain's no-code UX, built from scratch during this validation cycle, extended with comparative-study-aware handling not required in either prior domain, along with its fallback-detection fix and new pipeline-level binary validation, are verified working end-to-end against a real, company-derived employment dataset. This is now the third domain, after Finance and Health, to independently reproduce the same core pattern: a single dataset and model can be simultaneously fair on one protected attribute and unfair on another, and only auditing multiple attributes reveals the full picture. Hiring is now validated as a working reference implementation alongside Finance and Health, joining Justice and Education (already covered in earlier real-dataset validation work), leaving Business and Governance as the remaining FDK domains.